

Program 4: Demonstrate CI Integration by running Jenkins in a Docker container

Step 1: Create a directory and change to it

mkdir program5

cd program5

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab$ mkdir program5
```

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab$ cd program5
```

Step 2 : create dockerfile

nano Dockerfile

```
FROM jenkins/jenkins:lts

#switch to root user to install tools
USER root

#install docker inside jenkins container
RUN apt-get update && apt-get install -y docker.io

#switch back to jenkins user
USER jenkins
```

Step 3: Create docker-compose.yml

nano docker-compose.yml

```
version : '3.8'
services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

volumes:
  jenkins_home:
```

Step 4: build the container

docker compose build

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/prgm5$ docker compose build
WARN[0000] /home/1rv24mc040-haripriya/Devops_lab/prgm5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it
avoid potential confusion
[+] Building 7.0s (9/9) FINISHED
=> [internal] load local bake definitions                                0.0s
=> => reading from stdin 539B                                          0.0s
=> [internal] load build definition from Dockerfile                    0.0s
=> => transferring dockerfile: 249B                                    0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:lts        6.9s
=> [auth] jenkins/jenkins:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                       0.0s
=> => transferring context: 2B                                         0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:lts@sha256:7b1c378278279c8688efd 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io    0.0s
=> exporting to image                                                 0.0s
=> => exporting layers                                                0.0s
=> => writing image sha256:f5df4847ab9add46eb89d97e1eb3b1a3b166db5b950bb 0.0s
=> => naming to docker.io/library/prgm5-jenkins                      0.0s
=> resolving provenance for metadata file                             0.0s
[+] Building 1/1
✓ prgm5-jenkins Built 0.0s
```

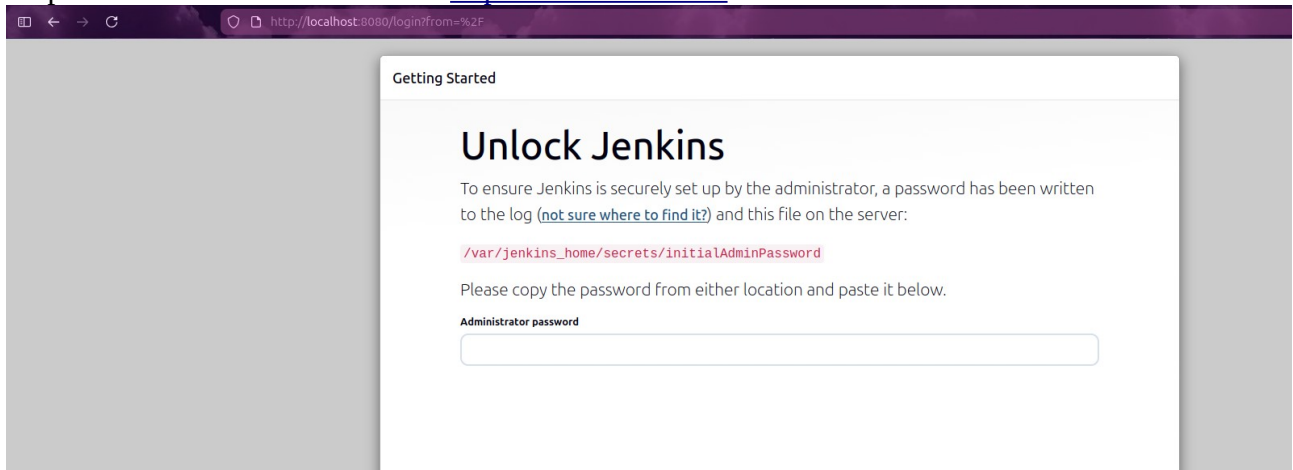
Step 5: up the image so that it runs

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/prgm5$ docker compose up -d
WARN[0000] /home/1rv24mc040-haripriya/Devops_lab/prgm5/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 3/3
✓ Network prgm5_default Created 0.1s
✓ Volume prgm5_jenkins_home Created 0.0s
✓ Container jenkins-ct Started 0.4s
```

Step 6: List the images and check for the newly created image docker images or docker ps

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/prgm5$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
b1f698f4f55e  prgm5-jenkins "/usr/bin/tini -- /u..." 22 minutes ago Up 22 minutes 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp jenkins-ct
```

Step 7: Check in the browser with <https://localhost:8080>



Step 6: Remove the docker image of jenkins

docker image rm program5-jenkins:latest

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/prgm5$ docker image rm program5-jenkins:latest
Untagged: program5-jenkins:latest
Deleted: sha256:f5d2b400e37581f1e99b7b1436becced75a2e914db8a4d93bec201a8d6aa988d
```